Week 10 Revision Selected Questions

Worked Solutions

Based on `Sem 2 Week 10 - Revision.pdf`

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1. GreenGrid Energy

Let $x_A, \ldots, x_F \in \{0, 1\}$ where $x_i = 1$ if project i is selected.

(a) A suitable binary model is

$$\max Z = 14x_A + 17x_B + 11x_C + 20x_D + 9x_E + 15x_F$$

subject to

$$6x_A + 8x_B + 5x_C + 9x_D + 4x_E + 7x_F \leq 21$$

$$x_D \leq x_B$$

$$x_A + x_F \leq 1$$

$$x_C + x_E \geq 1$$

$$x_i \in \{0, 1\} \quad \text{for all projects } i.$$

(b) The optimal solution is

$$x_B = x_D = x_E = 1, \qquad x_A = x_C = x_F = 0.$$

Total cost:

$$8 + 9 + 4 = 21$$

Total benefit:

$$17 + 20 + 9 = 46.$$

So the best project set is $\{B, D, E\}$ with maximum total annual benefit score 46.

(c) Binary variables are required because each project is indivisible. It is meaningful to choose or reject a project, but not to choose, for example, 0.4 of project D.

(d) Add the constraint

$$x_A + x_B + x_C + x_D + x_E + x_F = 4.$$

Under the budget and logic rules, there is no feasible solution selecting exactly four projects. So the revised model is infeasible.

(e) One realistic extension would be a multi-period capital budgeting model with separate yearly budgets and project timing decisions. This would need extra data on when costs are incurred, when benefits begin, and any project sequencing dependencies.

2. Hock-Brooke Automobiles

Corrected 15 Oct 2026 (erratum C1). The earlier version used a spreadsheet in which the *Delivered* and *Demand* tables were swapped (and the demand values shifted by one), and it charged overflow on the closing stock. This version uses the data as set, charges space on the stock held during the week (opening stock + Monday delivery), and reports expected costs over a 52-week horizon. The notebook `Hock-Brooke - warehouse simulation.ipynb` reproduces every figure.

New cars arrive on Monday and are collected on Friday or Saturday, so during the week the dealer must hold the leftover stock *plus* the delivery. For a capacity c, the weekly cost is $10c + 20\max(0,\ \text{stock} - c)$; sales are $\min(\text{stock}, \text{demand})$ and unsold cars carry over.

(a) For capacity 9, an illustrative 12-week simulation (random draws for deliveries and demand) is:

Week	Opening	Delivered	Stock held	Demand	Sales	Closing	Overflow	Cost
1	0	9	9	6	6	3	0	90
2	3	8	11	7	7	4	2	130
3	4	9	13	10	10	3	4	170
4	3	10	13	7	7	6	4	170
5	6	9	15	6	6	9	6	210
6	9	8	17	8	8	9	8	250
7	9	7	16	7	7	9	7	230
8	9	9	18	8	8	10	9	270
9	10	8	18	9	9	9	9	270
10	9	10	19	6	6	13	10	290
11	13	10	23	10	10	13	14	370
12	13	9	22	7	7	15	13	350

This run costs 2800 over 12 weeks, i.e. $2800/12 \approx 233.33$ per week. Overflow grows because the mean delivery (8.80 cars) exceeds the mean demand (8.65): unsold stock tends to build up by about 0.15 cars a week, so there is no stable long-run average and the answer depends on the horizon. Over one year (52 weeks, starting empty) the *expected* average weekly cost for capacity 9 is

$$\approx 243.78 \text{ per week}$$

(exact Markov-chain calculation; 2,000 simulated years give ≈ 242.5).

(b) Expected average weekly cost over 52 weeks, starting with an empty warehouse:

Capacity	Average Weekly Cost
6	272.57
7	262.57
8	252.84
9	243.78
10	236.48
11	230.81
12	226.54
13	223.64
14	222.01
15	221.56 **(lowest)**
16	222.22
17	223.89
18	226.49

So capacity 9 is not the best choice. Over a year the lowest expected cost is at about **15 spaces** (≈ 221.56 per week), with 14 and 16 within £1. Over a shorter 12-week horizon the best is 11 spaces (≈ 137.85 per week). Because stock keeps building up, the dealer should also review the order quantity, not only the warehouse size.

Note: other correctly executed simulations will give different short-run tables; the expected values above do not depend on the random numbers.

3. University Library Help Desk

(a) Main components:

Entities: students.

Resource: advisers.

Events: arrival, service start, service completion.

Stochastic inputs: interarrival times and service times.

Performance measures: number served, average wait, maximum wait and the percentage waiting less than 8 minutes.

(b) One valid baseline run gives:

Measure	Result
Students served	41
Average wait	1.17 minutes
Maximum wait	10.32 minutes
% waiting less than 8 minutes	95.12%

(c) Peak-period results:

Advisers	Served	Avg wait	Max wait	% below 8 min
2	70	1.88	12.99	95.71%
3	73	0.32	3.22	100.00%
4	70	0.08	2.23	100.00%

The target is 97% waiting less than 8 minutes. In this one run (seed 24), 3 advisers meet the target. But seed 24 is a light day: about 70 students are served, against 90 expected arrivals (15×6). A single run is not enough to choose a staffing level (see part (e)). Over 200 replications (seeds 0–199):

Advisers	Mean % below 8 min	Days meeting 97%
2	54.9%	1%
3	94.0%	52%
4	99.3%	92%

The Erlang C formula for the M/M/c queue agrees: 92.0% for 3 advisers and 99.1% for 4. The minimum staffing level that meets the target is therefore **4 advisers**.

Corrected 15 Oct 2026 (erratum C17). The earlier version answered 3 advisers, based only on the seed-24 run above. With replications, 3 advisers average about 94% and meet the target on only about half the days, so 4 are needed. The notebook `University library - simulation (SimPy).ipynb` reproduces every figure.

(d) A `PriorityResource` can be used instead of an ordinary `Resource`. Each arrival is labelled priority or standard, for example with probability 0.10 of being priority. Priority users would request service with a smaller priority number, such as `priority=0`, while standard users might request with `priority=1`. This would generally reduce waits for priority users and increase waits for standard users.

(e) A single run is not enough because simulation output is affected by random variation. Replications with different seeds give more stable estimates and allow the calculation of confidence intervals. If the system were modelled as an ongoing operation rather than a fixed 360-minute session, a warm-up period could also be used to reduce start-up bias.

4. F.B. Knight Deliveries

Because the costs are stated per 10 items, it is convenient to model the variables in tens of items. Then warehouse supplies are 10, 10 and 10, while shop demands are 7, 8 and 15.

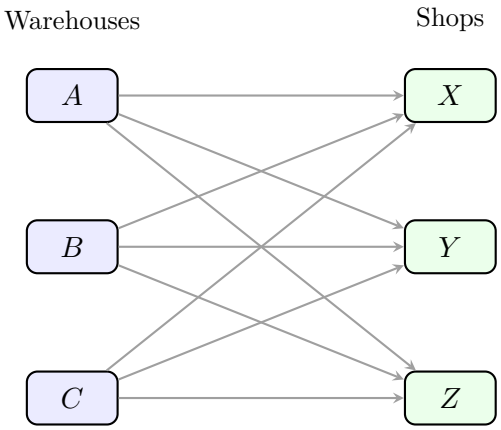
(a) Let

$$x_{ij} = \text{number of tens of items shipped from warehouse } i \text{ to shop } j$$

for

$$i \in \{A, B, C\}, \qquad j \in \{X, Y, Z\}.$$

The direct transportation network is:



The direct-shipping transportation model is

$$\min Z = 8x_{AX} + 7x_{AY} + 5x_{AZ} + 12x_{BX} + 3x_{BY} + 9x_{BZ} + 10x_{CX} + 11x_{CY} + 7x_{CZ}$$

subject to warehouse supply constraints

$$x_{AX} + x_{AY} + x_{AZ} = 10$$

$$x_{BX} + x_{BY} + x_{BZ} = 10$$

$$x_{CX} + x_{CY} + x_{CZ} = 10$$

shop demand constraints

$$x_{AX} + x_{BX} + x_{CX} = 7$$

$$x_{AY} + x_{BY} + x_{CY} = 8$$

$$x_{AZ} + x_{BZ} + x_{CZ} = 15$$

and non-negativity:

$$x_{ij} \geq 0.$$

One optimal direct-shipping plan is:

Route	Quantity (tens of items)	Cost per 10 items
$A \to Z$	10	5
$B \to X$	2	12
$B \to Y$	8	3
$C \to X$	5	10
$C \to Z$	5	7

Total cost:

$$10(5) + 2(12) + 8(3) + 5(10) + 5(7) = 183.$$

(b) With the restrictions

$$x_{BY} = 0$$

and

$$x_{AY} \geq 0.5$$

the same arc-variable model is modified by adding

$$x_{BY} = 0$$

and

$$x_{AY} \geq 0.5.$$

In tens-of-items units, one optimal revised plan is:

Route	Quantity (tens of items)
$A \to Y$	8
$A \to Z$	2
$B \to X$	7
$B \to Z$	3
$C \to Z$	10

Revised total cost:

$$8(7) + 2(5) + 7(12) + 3(9) + 10(7) = 247.$$

So the extra restrictions increase the cost from 183 to 247.

(c) If direct shipping is not allowed and the intermediary locations are used, define arc variables

$$y_{ik} = \text{flow from warehouse } i \text{ to intermediary } k$$

for

$$i \in \{A, B, C\}, \qquad k \in \{D_1, D_2\}$$

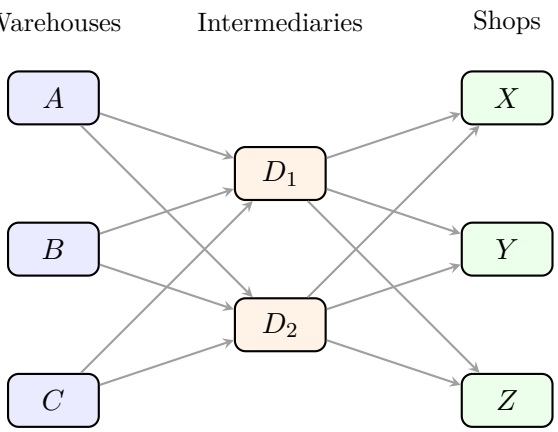
and

$$z_{kj} = \text{flow from intermediary } k \text{ to shop } j$$

for

$$k \in \{D_1, D_2\}, \qquad j \in \{X, Y, Z\}.$$

The transhipment network is:



The transhipment model is

$$\min Z = 4y_{AD_1}+5y_{AD_2}+3y_{BD_1}+2y_{BD_2}+6y_{CD_1}+4y_{CD_2}+4z_{D_1X}+5z_{D_1Y}+2z_{D_1Z}+2z_{D_2X}+4z_{D_2Y}+4z_{D_2Z}$$

subject to supply constraints at the warehouses

$$y_{AD_1} + y_{AD_2} = 10$$

$$y_{BD_1} + y_{BD_2} = 10$$

$$y_{CD_1} + y_{CD_2} = 10$$

demand constraints at the shops

$$z_{D_1X} + z_{D_2X} = 7$$

$$z_{D_1Y} + z_{D_2Y} = 8$$

$$z_{D_1Z} + z_{D_2Z} = 15$$

and flow-balance constraints at the intermediary nodes

$$y_{AD_1} + y_{BD_1} + y_{CD_1} = z_{D_1X} + z_{D_1Y} + z_{D_1Z}$$

$$y_{AD_2} + y_{BD_2} + y_{CD_2} = z_{D_2X} + z_{D_2Y} + z_{D_2Z}.$$

All arc flows satisfy

$$y_{ik} \geq 0, \qquad z_{kj} \geq 0.$$

If only one intermediary is used, then the variables on the unused node are omitted from the model. The three possibilities are:

Intermediary choice	Total cost
D_1 only	228
D_2 only	216
Both D_1 and D_2	181

So both intermediary locations should be used.

One optimal transhipment plan is:

Route	Quantity (tens of items)
$A \to D_1$	10
$B \to D_1$	5
$B \to D_2$	5
$C \to D_2$	10
$D_1 \to Z$	15
$D_2 \to X$	7
$D_2 \to Y$	8

Total cost:

$$10(4) + 5(3) + 5(2) + 10(4) + 15(2) + 7(2) + 8(4) = 181.$$

This is slightly cheaper than the best direct-shipping plan.

5. Forecasting at N&F

(a) The quarterly data show a clear upward trend over time. Using centred 4-quarter moving averages, the smoothed values where available are:

Period	Centred 4-quarter moving average
3	1020.250
4	1030.500
5	1040.750
6	1050.625
7	1060.500
8	1070.375
9	1080.375
10	1090.625
11	1100.875
12	1111.125
13	1121.250
14	1131.000

(b) Estimated seasonal variations:

Additive seasonal values:

$$\text{Spring} = 12.48, \quad \text{Summer} = 15.85, \quad \text{Autumn} = -18.27, \quad \text{Winter} = -10.06.$$

Multiplicative seasonal indices:

$$\text{Spring} = 1.0117, \quad \text{Summer} = 1.0147, \quad \text{Autumn} = 0.9829, \quad \text{Winter} = 0.9907.$$

(c) A linear trend fitted to the centred moving averages is approximately

$$\hat{T}_t = 990.12 + 10.07t.$$

Forecasts for next year are therefore:

Period	Season	Additive Forecast	Multiplicative Forecast
17	Spring	1173.74	1174.83
18	Summer	1187.18	1188.53
19	Autumn	1163.12	1161.18
20	Winter	1181.40	1180.42

(d) Both models give very similar forecasts. However, the additive model is slightly easier to justify because the seasonal effects appear roughly constant in size over time, rather than increasing proportionally with the level of the series. So an additive model would be a reasonable recommendation here, although either model would give very similar practical forecasts.

End of Solutions